End-to-End Sales Performance & Business Insights System

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1. Problem Statement

Businesses often face situations where sales are increasing, but profits are not growing at the same rate. This makes it difficult to understand whether the business is truly performing well or not.

2. Project Objective

In real-world business scenarios, companies deal with large amounts of sales data coming from different regions, customers, and products. However, raw data alone does not provide value unless it is properly analyzed.

Some common challenges businesses face:

- High sales but low profit
- Inefficient discount strategies
- Underperforming regions or products
- Lack of visibility into customer behavior
- Delays in order delivery affecting operations

This project aims to solve these challenges by transforming raw data into structured insights.

Dataset Summary

(Source: *Kaggle datasets; SuperStore Sales Dataset by Aditi Saxena*)

The dataset contained over 50,000 sales records, including information about orders, customers, products, regions, and profits. I worked on cleaning, analyzing, and visualizing this data step-by-step using different tools.

Tools Used

- **Data Engineering:** Python (Pandas, NumPy), SQLAlchemy
- **Database Management:** MySQL (Relational Schema, Window Functions, Ntile, Subqueries, CTEs)
- **Exploratory Data Analysis & Data Validation:** Excel
- **Business Intelligence:** Power BI (for dashboarding)

Phase 1: Data Cleaning & Feature Engineering

The goal of this phase was to turn raw, messy CSV file into a high-quality "Master Dataset”.

1. Data Ingestion

- I developed a Python script using the SQLAlchemy and logging libraries. Instead of a simple copy-paste, I implemented logging to track the success or failure of every row inserted into MySQL. This ensures the pipeline is "Production-Ready."
- To ensure seamless compatibility across the full data stack (**MySQL, Python/Pandas, Excel and Power BI**), I implemented a ‘Standardized Naming Convention’ using **snake_case**.
- Raw data was first ingested into **SQL** for structured storage and initial cleaning. The processed table were then exported to **Python (Pandas)** for advanced feature engineering.

2. Data Cleaning

➢ ***Duplicates Handling***

- I ran a check for 100% identical rows across every single column. The result was zero. This confirms that every row in my dataset represents a unique part of a business transaction, and there is no 'garbage' or double-entry data to remove.

➢ ***Data Inconsistencies Audit***

- Converted date columns into proper datetime format.
- Sales column was in text, converted to float.
- Standardized text fields for consistency.
- Checked logical errors such as shipping date cannot preceed order date.
- Checked Price errors. Found 1 row where sales is equal to 0. Filtered out that row to maintain data consistency.

```
df.info()
```
✓ 0.0s

```
<class 'pandas.core.frame.DataFrame'>
Index: 51289 entries, 0 to 51289
Data columns (total 21 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   order_id        51289 non-null  object
 1   order_date      51289 non-null  datetime64[ns]
 2   ship_date       51289 non-null  datetime64[ns]
 3   ship_mode       51289 non-null  object
 4   customer_name   51289 non-null  object
 5   segment         51289 non-null  object
 6   state           51289 non-null  object
 7   country         51289 non-null  object
 8   market          51289 non-null  object
 9   region          51289 non-null  object
 10  product_id      51289 non-null  object
 11  category        51289 non-null  object
 12  sub_category    51289 non-null  object
 13  product_name    51289 non-null  object
 14  sales           51289 non-null  float64
 15  quantity        51289 non-null  int64
 16  discount        51289 non-null  float64
 17  profit          51289 non-null  float64
 18  shipping_cost   51289 non-null  float64
 19  order_priority  51289 non-null  object
 20  year            51289 non-null  int64
dtypes: datetime64[ns](2), float64(4), int64(2), object(13)
memory usage: 8.6+ MB
```

3. Feature Engineering

To provide deeper insights, **3+ custom columns (features)** were engineered.

- **profit_margin:** Total Profit is a volume metric but Profit Margin is performance metric. It allows us to identify which categories are most cost-effective and which ones might be dragging down revenue due to high shippings cost or heavy discounting. In general it is difference between “how much we made” vs. “how well we are performing”.
- **delivery_days:** We create delivery_days to measure operational efficiency. It is a key performance indicator for logistic efficiency. It allows us to perform comparative analysis across different shipping modes and region. Long delivery time are the #1 reason for bad reviews. By calculating this, we can find where the bottlenecks are. For example if first class shipping mode taking 1-2 days for delivery then they are doing great. If standard class taking 7 to 8 days then they are too slow, they need to optimize the warehouse location.
- **high_discount:** A high discount is often the hidden reason why a product has a high slaes volume but a negative profit. By creating this feature we are telling to the business that we are selling a lot, but we are giving away too much for free or at much less price. By creating this we can see that, are customers buying only when high discounts are applied?

	order_id	order_date	ship_date	ship_mode	customer_name	segment	state	country	market	region	...	profit	shipping_cost	order_priority	year	delivery_days	profit_margin	high_discount	order_year	order_month	order_month_name
0	AG-2011-2040	2011-01-01	2011-01-06	Standard Class	Toby Braunhardt	Consumer	Constantine	Algeria	Africa	Africa	...	106.1400	35.46	Medium	2011	5	26.014706	0	2011	1	January
1	IN-2011-47883	2011-01-01	2011-01-08	Standard Class	Joseph Holt	Consumer	New South Wales	Australia	APAC	Oceania	...	36.0360	9.72	Medium	2011	7	30.030000	0	2011	1	January
2	HU-2011-1220	2011-01-01	2011-01-05	Second Class	Annie Thurman	Consumer	Budapest	Hungary	EMEA	EMEA	...	29.6400	8.17	High	2011	4	44.909091	0	2011	1	January
3	IT-2011-3647632	2011-01-01	2011-01-05	Second Class	Eugene Moren	Home Office	Stockholm	Sweden	EU	North	...	-26.0550	4.82	High	2011	4	-57.900000	1	2011	1	January
4	IN-2011-47883	2011-01-01	2011-01-08	Standard Class	Joseph Holt	Consumer	New South Wales	Australia	APAC	Oceania	...	37.7700	4.70	Medium	2011	7	33.131579	0	2011	1	January
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
51285	CA-2014-115427	2014-12-31	2015-01-04	Standard Class	Erica Bern	Corporate	California	United States	US	West	...	4.5188	0.89	Medium	2014	4	32.277143	0	2014	12	December
51286	MO-2014-2560	2014-12-31	2015-01-05	Standard Class	Liz Preis	Consumer	Souss-Massa-Draâ	Morocco	Africa	Africa	...	0.4200	0.49	Medium	2014	5	10.500000	0	2014	12	December
51287	MX-2014-110527	2014-12-31	2015-01-02	Second Class	Charlotte Melton	Consumer	Managua	Nicaragua	LATAM	Central	...	12.3600	0.35	Medium	2014	2	47.538462	0	2014	12	December
51288	MX-2014-114783	2014-12-31	2015-01-06	Standard Class	Tamara Dahlen	Consumer	Chihuahua	Mexico	LATAM	North	...	0.5600	0.20	Medium	2014	6	8.000000	0	2014	12	December
51289	CA-2014-156720	2014-12-31	2015-01-04	Standard Class	Jill Matthias	Consumer	Colorado	United States	US	West	...	-0.6048	0.17	Medium	2014	4	-20.160000	0	2014	12	December

51289 rows × 27 columns

Phase 2: SQL – Querying and Business Analysis

I loaded our clean dataset into a MySQL database to solve business questions. I used **Window Functions (RANK())** , **NTILE & CTEs** to extract deep insights.

- ***Total Sales, Profit, Quantity***

	total_sales	total_quantity	total_profit
▶	12642905	178311	1469035.93

  - Our business is very large in terms of sales, making 1.2 crore. However our profit margin is around 11.5%. This means that for every 100 rupees we are making only about 11-12 rupees as profit. Rest goes into logistic cost.

- ***Monthly Sales Trend.***

order_year	order_month_name	monthly_sales
2011	December	333941
2011	November	298499
2011	September	290230
2011	June	215214
2011	August	207570
2011	October	199070
2011	May	146762
2011	March	145726
2011	April	116927
2011	July	115518
2011	January	98902
2011	February	91152
2012	December	338245
2012	November	323512
2012	August	303158
2012	September	289390
2012	June	256181
2012	October	252942
2012	May	208370
2012	March	163092

- Our business follows a seasonal pattern. Sales start very slow in January and February but they grow steadily throughout the year. The peak season is November and December, where we make most money. This is likely end year sale or festive offers.

- ***Top 10 Products by Sales***

product_name	category	total_sales	percentage_share
Apple Smart Phone, Full Size	Technology	86936	0.69%
Cisco Smart Phone, Full Size	Technology	76441	0.6%
Motorola Smart Phone, Full Size	Technology	73159	0.58%
Nokia Smart Phone, Full Size	Technology	71904	0.57%
Canon imageCLASS 2200 Advanced Copier	Technology	61600	0.49%
Hon Executive Leather Armchair, Adjustable	Furniture	58200	0.46%
Office Star Executive Leather Armchair, Adjustable	Furniture	50667	0.4%
Harbour Creations Executive Leather Armchair, Adjustable	Furniture	50120	0.4%
Samsung Smart Phone, Cordless	Technology	48654	0.38%
Nokia Smart Phone, with Caller ID	Technology	47880	0.38%

- Our top revenue comes from premium technology. Specifically smart phones like Apple and Cisco. Technolgy category is the highest profit driver for our business.

- ***Loss Making Products***

	product_name	total_profit
▶	Cubify CubeX 3D Printer Double Head Print	-8879.970428466797
	Lexmark MX611dhe Monochrome Laser Printer	-4589.972961425781
	Motorola Smart Phone, Cordless	-4447.037979125977
	Cubify CubeX 3D Printer Triple Head Print	-3839.990478515625
	Bevis Round Table, Adjustable Height	-3649.8939514160156
	Bevis Computer Table, Fully Assembled	-3509.5637817382812
	Rogers Lockers, Blue	-2893.490818977356
	Chromcraft Bull-Nose Wood Oval Conference Tables & Bases	-2876.115608215332
	Bevis Wood Table, with Bottom Storage	-2782.587965965271
	Lesro Training Table, Rectangular	-2581.283447265625

  - We have several products where the total profit is negative. This means even though we are selling them, we are actually losing money on every sale. This could be because of very high shipping costs, too much discounting or high returns rate from customers.

- ***Region-wise Performance***

	region	total_sales	total_profit
▶	Central	2822399	311405.09
	South	1600960	140355.77
	North	1248192	194597.95
	Oceania	1100207	121666.64
	Southeast Asia	884438	17852.33
	North Asia	848349	165578.42
	EMEA	806184	43897.97
	Africa	783776	88871.63
	Central Asia	752839	132480.19
	West	725514	108418.45
	East	678834	91522.78
	Caribbean	324281	34571.32
	Canada	66932	17817.39

  - Our business is very sucessful in Central Region, where we have our highest sale of 28 Lakhs. However our Canada Region is very small with only 66 thousand in sales. This shows our brand is much stronger in Central markets than in North America.

- ***Discount Impact Analysis***

high_discount	avg_profit_margin	total_sales
0	23.39%	10712663
1	-61.15%	1930242

  - Our current discount strategy is failling, when we give more than a 20% discount, our profit margin falls to -61% meaning we lose 61 rupees for every 100 rupees of sales. Even with high discount total sales do not increase. This means customers are not attracted by deep discounts on specific products.

- ***Top Product in Each Category***

category	product_name	total_sales	rnk
Furniture	Hon Executive Leather Armchair, Adjustable	58200	1
Office Supplies	Eldon File Cart, Single Width	34392	1
Technology	Apple Smart Phone, Full Size	86936	1

  - Our business has a strong market leader in every category. In Furniture, people wants comfort products like Armchairs. In Office Supplies, they want organization files. In Technology, they want premium brands like Apple. These products have the highest sales.

- ***Customer Segmentation (High Value Customers)***

customer_name	total_sales	priority
Tom Ashbrook	40489	(Top 25%) Top-Tier
Tamara Chand	37453	(Top 25%) Top-Tier
Greg Tran	35552	(Top 25%) Top-Tier
Christopher Conant	35187	(Top 25%) Top-Tier
Sean Miller	35170	(Top 25%) Top-Tier
Bart Watters	32315	(Top 25%) Top-Tier
Natalie Fritzler	31778	(Top 25%) Top-Tier
Fred Hopkins	30404	(Top 25%) Top-Tier
Jane Waco	30288	(Top 25%) Top-Tier
Hunter Lopez	30246	(Top 25%) Top-Tier
Sanjit Engle	30143	(Top 25%) Top-Tier
Penelope Sewall	30083	(Top 25%) Top-Tier
Peter Fuller	29876	(Top 25%) Top-Tier
Muhammed Yedwab	29644	(Top 25%) Top-Tier
Raymond Buch	29601	(Top 25%) Top-Tier

- The top 25% (Top-Tier) are our most important customers because they provide the highest revenue. On the other hand, the bottom 25% (Low-Value) may only buy once or twice. We need to treat these groups seperately to grow sales.

- ***Operational Efficiency***

region	ship_mode	avg_delivery_days	total_sales	avg_profit_margin
North	Standard Class	5.0593	749478	12.88
West	Standard Class	5.0579	406787	21.53
Central	Standard Class	5.0375	1722593	5.71
Caribbean	Standard Class	5.0040	186649	8.65
North Asia	Standard Class	5.0000	517898	17.91
Canada	Standard Class	4.9845	30040	24.32
South	Standard Class	4.9754	927160	5.41
Oceania	Standard Class	4.9731	675471	8.94
Africa	Standard Class	4.9714	438521	-15.15
East	Standard Class	4.9701	405341	17.01
EMEA	Standard Class	4.9671	474979	-12.09
Southeas...	Standard Class	4.9512	560996	-8.42
Central Asia	Standard Class	4.9052	482976	14.5
Southeas...	Second Class	3.4073	142094	-5.38
Oceania	Second Class	3.2824	230999	8.15
Central	Second Class	3.2753	560801	6.83
East	Second Class	3.2660	116563	17.88
Africa	Second Class	3.2550	186330	-10.07

- Our data shows that Standard Class shipping takes the longest (usually 5-6 days) but has the highest sales volume. However Same day or First Class shipping is must faster but often has a lower profit margin because shipping costs are so high. We are sacrificing profit for speed in some regions.

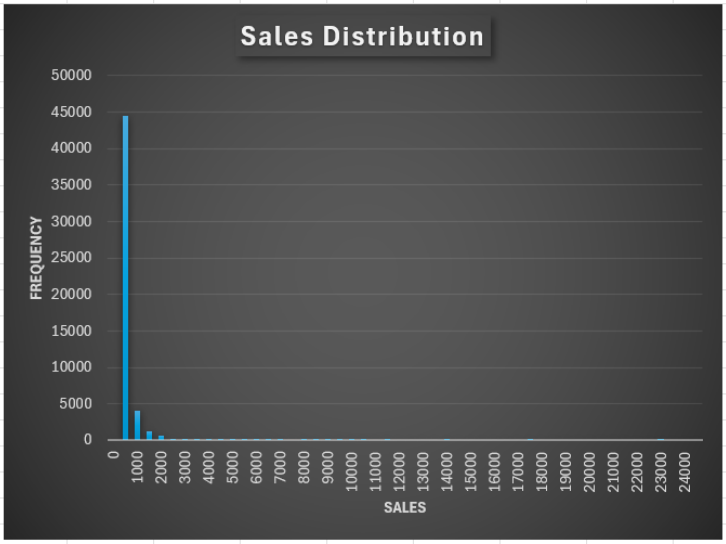
Phase 3: Excel Quick Analysis & Validation

This phase was about "Seeing the Data" and validating our findings. Using Excel statistical analysis, I performed an **Exploratory Data Analysis (EDA)** to understand sales performance.

1. Statistical Analysis

a) **Distribution Plots:** I checked the spread of Sales and Profit.

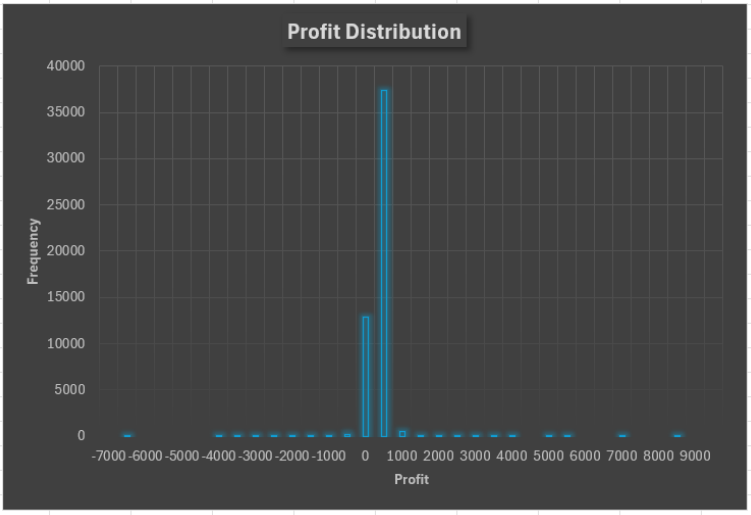
Sales	Frequency
0	0
500	44547
1000	4120
1500	1299
2000	616
2500	276
3000	196
3500	100
4000	45
4500	29
5000	17
5500	16
6000	9
6500	2
7000	3
7500	0
8000	2
8500	3
9000	1
9500	2
10000	1



Key Insights: Sales

- Our sales are **Right-Skewed** (Positively Skewed). This means most of our business relies on **High-Volume, Low-Price** sales. Nearly **90%** of our orders are for small amounts (less than 500). So we need high traffic to make money.

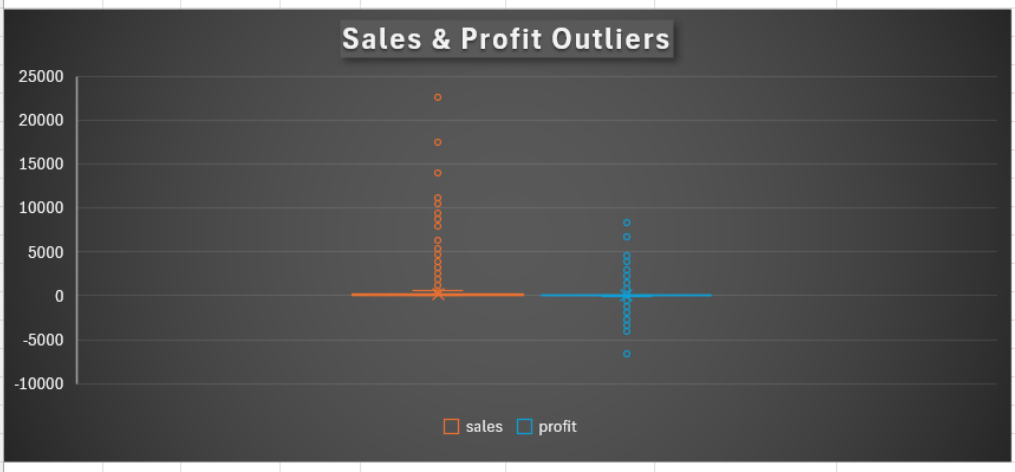
Profit	Frequency
-7000	0
-6500	1
-6000	0
-5500	0
-5000	0
-4500	0
-4000	1
-3500	2
-3000	3
-2500	3
-2000	3
-1500	21
-1000	44
-500	200
0	12932
500	37401
1000	530
1500	100
2000	23
2500	11
3000	6



Key Insights: Profit

- While most of our orders make a small profit, I found a 'Risk Zone' on the left side of my histogram. There are more than **200 transactions** where we lost significant money. This matches my SQL finding that high discounts are destroying our profit on certain items.

b) **Outlier Detection**



Outliers	Q1	Q3	IQR	Upper Bound	Lower Bound	Total Ouliers: Low	Total Ouliers: High
Sales	31	251	220	581	-299	0	5655
Profit	0	36.81	36.81	92.03	-55.22	3535.00	6230

Key Insights: Outliers

- **Sales (High):** These 5,655 orders are our 'Revenue Drivers.' Even though they are only about 10% of our total orders, they likely contribute more than 50% of our total revenue. These are our premium products.
- **Profit (High):** These are our 'High-Value Products.' We have over 6,000+ orders that are highly profitable. In SQL analysis I found that these products mostly belongs to 'Technology' category. For more profit, we should sell more products from this category.
- **Profit (Low):** We have over 3,500 orders where we are not just making a small profit but we are losing significant money. These are critical losses likely because a massive discount combined with a high shipping cost.

c) **Correlation Heatmap**

	sales	quantity	discount	profit	shipping_cost	delivery_days	profit_margin	high_discount
sales	1.00							
quantity	0.31	1.00						
discount	-0.09	-0.02	1.00					
profit	0.49	0.10	-0.32	1.00				
shipping_cost	0.77	0.27	-0.08	0.36	1.00			
delivery_days	0.00	0.00	0.00	0.00	-0.14	1.00		
profit_margin	0.07	0.05	-0.85	0.36	0.07	0.00	1.00	
high_discount	-0.08	-0.02	0.90	-0.31	-0.08	0.00	-0.75	1.00

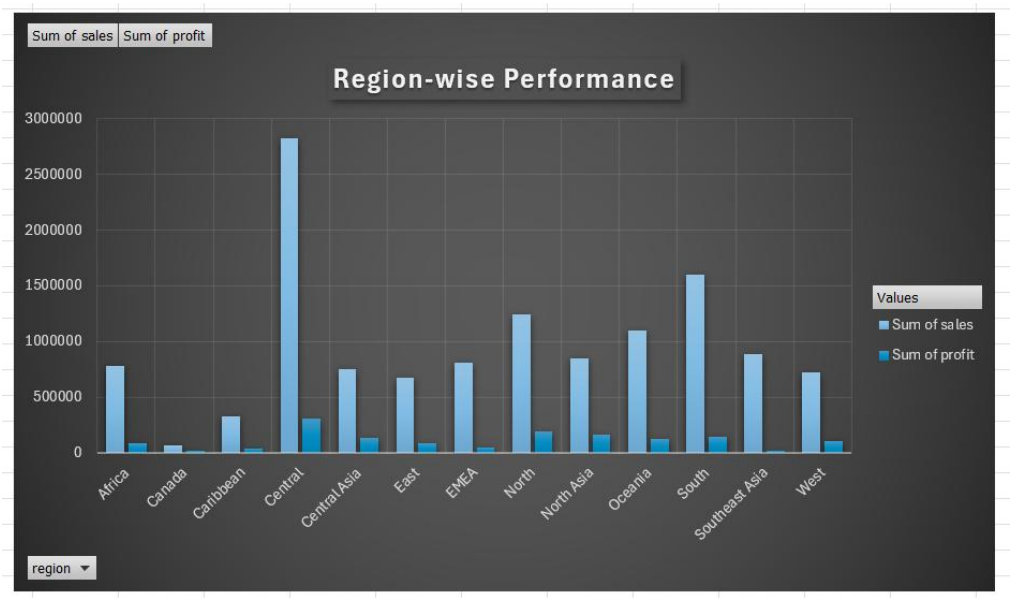
Key Insights: Correlation Heatmap

- **High Discount vs. Profit Margin (-0.75):** It proves that giving high discounts destroys our profit margin.
- **High Discount vs. Sales (-0.08):** The correlation is almost Zero. This proves that giving a high discount does not help us sell more products. The customers aren't buying more just because of the discount.
- **Profit vs. Profit Margin (0.36):** There is a positive link here, but it's not super strong. It shows that a high-profit item doesn't always have a high-profit percentage.

2. Validation

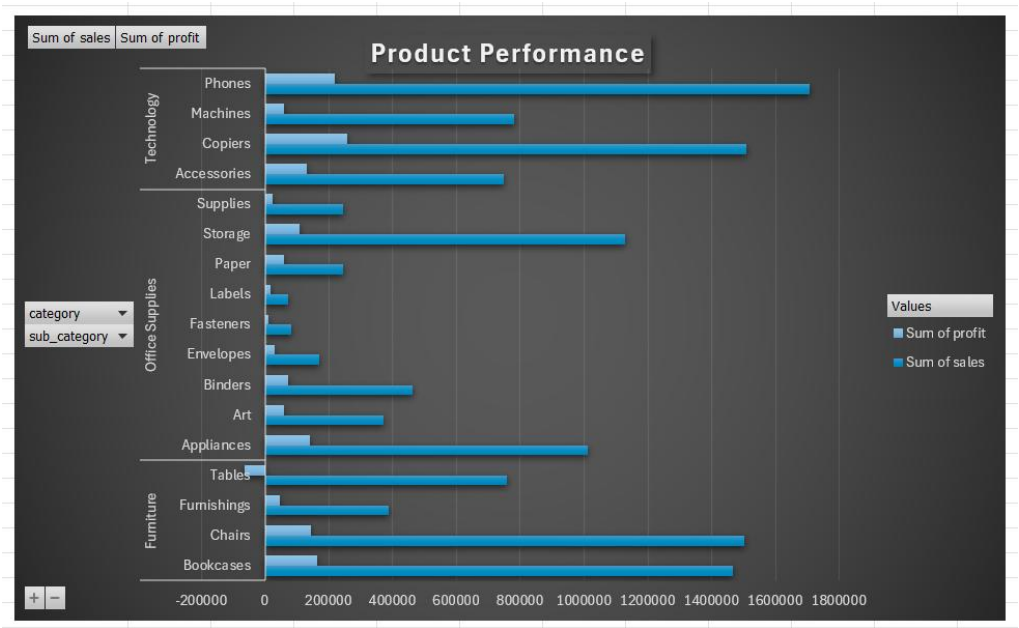
- Created pivot tables and charts for sales and profit analysis
- Analyzed regional and product performance
- Verified key metrics calculated in SQL

- ***Region-wise Performance***



  - **Recommendation:**
    - We should invest more in Central Region to stay the leader.
    - We should look for a new warehouse location in the Canada region to lower these shipping costs. If we reduce shipping costs by even 5%, our total profit will jump significantly.

- ***Product Performance***

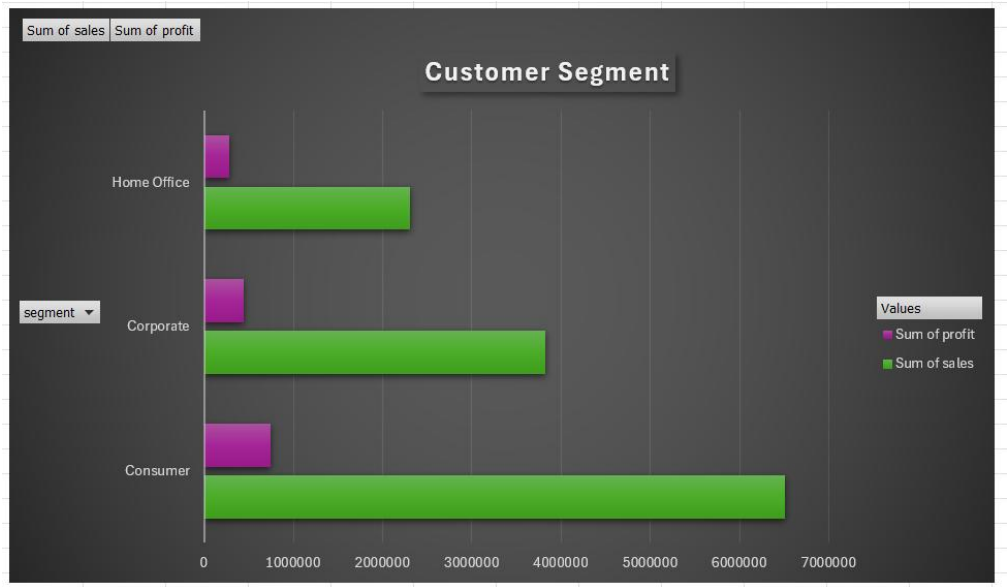


- **Recommendation:**
  - To increase sales of other products, we should use bundling strategies for example if someone buys executive chair we should recommend another product from same category with small discount.
  - We should stop selling tables in regions where shipping costs are high. Also, we should remove 'Tables' from all 'Sales/Discount' events. Tables should only be sold at full price to cover their high logistics costs.

- ***Monthly Sales Trend***

- **Recommendation:**
  - Since peak sale occurs in November, we should prepare for it by increasing stock by the end of October.
  - When sales are lowest in January and February we should run special discounts for steady money and plan for next peak sale.

- ***Customer Segment***



- **Recommendation:**
  - We should create a Loyalty Program specifically for the Top 25% of our Consumers. Instead of giving them deep discounts, we should give them Priority Shipping or Early Access to new Technology products.
  - Encourage Home Office buyers to place one large order per month instead of four small ones. This reduces our logistics costs.

- ***Discount vs Profit***

- **Recommendation:**
  - We must stop deep discounting on the product nobady wants. Instead of just discounting we should stop buying these products in the future.
  - Since sales are high at lower discount, it shows that our customers value the product quality and demand more than low price.

- ***Delivery Performance***

- **Recommendation:**
  - Since standard class is profitable, we should encourage more customers to use it by giving little discount for waiting longer.
  - For same day or first class delivery we must increase shipping fees to deal with lower profit margin. We should not lose money just to deliver a product in 24 hours.

Phase 4: Dashboard & Storytelling

Finally, I built an 3-page interactive dashboard in Power BI to present insights visually.

❖ Page 1: Sales Overview

- **KPI Cards:** Instant view of Sales, Quantity, Profit, & Profit Margin.
- **Slicers:** Date, Region.
- **Line Chart:** Sales trend by Year, Quarter, & Month.
- **Map Visual:** Sales contribution by Region.
- **Donut & Funnel Chart:** Sales by segment and customer tier.
- **Dynamic Insights:** Provides instant context.

❖ Page 2: Product Performance

- **KPI Cards:** Instant view of Average Discount and Shipping Cost.
- **Slicers:** Region, Ship Mode, & Category.
- **Bar Chart:** Top products vs loss making products.
- **Column Chart:** Category-wise sales and profit.
- **Dynamic Insights:** Provides instant context.

❖ Page 3: Customer & Discount

- **Gauges:** Average Discount and Profit Margin target.
- **Slicers:** Segment & Priority.
- **Scatter Plot:** Discount vs. profit.
- **Table Visual:** Top 10 customers by sales.
- **Donut Chart:** Sales contribution by customer tier.
- **Dynamic Insights:** Provides instant context.

Final Project Conclusion & Final Recommendations

We are running a business with **1.2 Crore** in sales, but we are leaking money. The problem is we **lost 64,000** just by selling tables. The reason is we are giving too much **discount (above 20%)** and then paying for expensive shippings which results; more table we sell, more money we actually lose.

Final Recommendations:

- Never give more than a 15% discount on Furniture.
- Make Customers Pay for Shipping on Big Items.
- Focus on "Quality" Customers, Not just "Big" Customers.
- Encourage more customers to use “Standard shipping” by giving little discount for waiting longer.